

Program: Identify Keyword, Identifier, or Comment

Algorithm

Step 1: Start the program.

Step 2: Read input from the user into a character array.

Step 3: If the input is `END`, stop the program.

Step 4: Call `check_comment`.

- Checks if input is a single-line comment, multi-line comment, or invalid comment.

- If input is a comment, print the type and go to Step 2.

Step 5: Call `check_keyword`.

- Checks if input is one of: `int`, `char`, `if`, `while`, `for`.

- If input is a keyword, print it and go to Step 2.

Step 6: Call `check_identifier`.

- Checks if input starts with a letter or underscore and contains only letters, digits, or underscores.

- If valid, print identifier. Otherwise, print that input is not a keyword, identifier, or comment.

Step 7: Go back to Step 2 to read next input.

Step 8: End the program.

Program Code

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int check_comment(char input[]) {
    int i;
    if (input[0] == '/') {
        if (input[1] == '/') {
            printf("It is a single-line comment\n");
            return 1;
        }
        else if (input[1] == '*') {
            for (i = 2; input[i] != '\0' && i < 28; i++) {
```

```

        if (input[i] == '*' && input[i + 1] == '/') {
            printf("It is a multi-line comment\n");
            return 1;
        }
    }
    printf("It is not a valid multi-line comment (missing */)\n");
    return 1;
}
else {
    printf("It starts with / but is not a comment\n");
    return 1;
}
}
return 0;
}

int check_keyword(char input[]) {
    char keywords[5][10] = {"int", "char", "if", "while", "for"};
    int i;
    for (i = 0; i < 5; i++) {
        if (strcmp(input, keywords[i]) == 0) {
            printf("%s : Keyword\n", input);
            return 1;
        }
    }
    return 0;
}

int check_identifier(char input[]) {
    int i;
    if (!isalpha(input[0]) && input[0] != '_' ) {
        return 0;
    }
    for (i = 1; input[i] != '\0'; i++) {
        if (!isalnum(input[i]) && input[i] != '_' ) {
            return 0;
        }
    }
    printf("%s : Identifier\n", input);
    return 1;
}

int main(void) {
    char input[30];
    int i;

    printf("Enter input (type END to stop):\n");
    while (1) {
        printf("> ");
        fgets(input, sizeof(input), stdin);

```

```

    for (i = 0; input[i] != '\0'; i++) {
        if (input[i] == '\n') {
            input[i] = '\0';
            break;
        }
    }

    if (strcmp(input, "END") == 0) {
        break;
    }

    if (check_comment(input)) {
        continue;
    }
    if (check_keyword(input)) {
        continue;
    }
    if (check_identifier(input)) {
        continue;
    }

    printf("%s : Not a keyword, identifier, or comment\n", input);
}

return 0;
}

```